

Recursive Fibonacci Function

Language: **Python** | Prepared by **Aman Samriya** | 2026-05-13

Original Code

```
def fib(n):
    if n <= 1:
        return n
    return fib(n-1) + fib(n-2)
```

What this code does (Layman)

This function calculates a number from the Fibonacci sequence, where each number is the sum of the previous two numbers (0, 1, 1, 2, 3, 5, 8, ...).

Step-by-Step Explanation

- Function fib takes a number n as input
- If n is 0 or 1, return n directly (base case)
- Otherwise, call fib(n-1) and fib(n-2)
- Return the sum of those two recursive results

Real-World Analogy

Imagine asking two younger siblings to tell you a number, and they ask their younger siblings, until someone knows the answer directly.

Inputs & Outputs

Input	Output
n=0	0
n=1	1
n=5	5
n=10	55

What it Delivers

Computes the Nth Fibonacci number using simple recursion. Useful for teaching recursion concepts.

Edge Cases & Assumptions

- Negative numbers are not handled

- Very large n causes deep recursion and slowness
- Stack overflow risk for $n > \sim 990$

■ Strengths

- Very easy to read
- Mirrors mathematical definition
- Great for teaching recursion

■■ Weaknesses

- Exponential time complexity $O(2^n)$
- Recomputes same values many times
- Not suitable for production

■■ Suggested Improvements

- Use memoization with a cache
- Convert to iterative loop
- Add input validation

■ Final Summary

A clean but inefficient recursive Fibonacci implementation, ideal for learning but unsuitable for production.